Offshore rig inspection & repair fleet — policy technical report

Go2 quadruped + lift + UR3 arm (scrubber / sprayer / torque tool). Every number below is read from a file in the repository (`runs_sample/`, `policy/SCRUB.md`, `detection/RESULTS.md`, run summaries); *all results are simulation results* (Isaac Lab or CPU mocks) — nothing has run on hardware yet.

Policy	Conditioned on (observation)	Action / output	How trained / designed	Runs on · latency	Headline measured result
1 Rust scrub PPO v2 `Rig-Scrub-Go2UR3Lift-v2`	327-D: UR3 $q, \dot{q}$ (12); brush pos/vel in pipe frame (6); gap, $\sin\theta, \cos\theta$, axial (4); target−brush (3); contact force F, setpoint F^*, pipe radius (3); 9×9 rust patch + 9×9 sprayed patch under brush (162); 8×16 coarse rust map (128); nearest-rust vector + dist (3); rust left, unsprayed-contact flag (2); last action (4)	4-D @30 Hz: brush-target step in pipe frame ($\pm$1 cm/axis) + F^*=25$\pm$20 N; F^* clipped at 38 N outside the net; DLS IK → UR3 joint targets	PPO, MLP 512-256-128. v1: 2048 envs×150 it on GB10 (≈9 min); v2 = 3-stage fine-tune, 500 it. Reward: +50 Δremoved, +20 success, contact band 10–40 N, over-force $1.0\frac{F-40}{10} + 4\left(\frac{F-30}{10}\right)_+^2$, unsprayed, action-rate. DR: spring k 1.5–4 kN/m, abrasion 0.5–1.5, 1–4 patches, pipe yaw $\pm10°$, new patch mid-episode	GB10 edge; ONNX 17 μs p50 on 1 Arm core; Isaac 8.3 ms per 256-env batch	Isaac, 256 ep: **95.3%** success [92.0, 97.3], 7.05 s to clean, **1.9%** ep. >40 N (greedy 32.0%/46.5%, random 0%/68.4%). Rig geometry, 1026 ep: 97.6%, 1.0% >40 N (v1: 76.8%). Contact sensor in the loop, 512 ep: 91.6%, 0% >40 N, max 39.4 N. Force sweep: removal saturates at F^* ≈35 N → ≈20 N mean at the tip
2 Storm scrub v3 `policy/storm.py`	Same 327-D obs (drop-in ONNX); disturbance is in the dynamics only	same 4-D	v2 fine-tuned 150 it with storm level $L\sim U(0,1)$: gusts 10–30 N·L on arm links, deck sway 2–5 cm at 0.1–0.3 Hz, wet pipe abrasion×(1−0.35L)	GB10 edge, as v2	≈1024 ep each, 100% success throughout. >40 N: v2 calm 0.5%, v2 L=1 12.0% (max 274 N), v3 L=1 9.0% (max 69 N). Orchestrator gust gate (retract >12 N, resume <6 N) halves v2 to 6.6%
3 Walking `Rig-Walk-Go2UR3Lift-v0`	236-D: base lin/ang vel, gravity, velocity command (12); 12 leg q, $\dot{q}$, last action (36); 187-point height scan; payload α (arm stowed→deployed)	12 leg joint position targets	Isaac Lab Go2 rough PPO, 4096 envs×1500 it, 99 min, L40S. Deck, grating, thresholds, plates, 3–12 cm steps; DR mass, CoM, pushes, deck tilt, arm deploy	GB10 edge (ONNX); *not measured*	35 km eval: vel. error 0.064 m/s, yaw 0.062 rad/s, 1.94 falls/km (worst: deck 2.6). Patrol 65 m route to the pipe: 16/16 reach in 67.8 s, final lateral error 4.4 cm
4 Can pick-up `Rig-PickCan-Go2UR3Lift-v0`	42-D, no pixels: depth ray-cast (32×32, mast) → cylinder fit (centre, axis, radius, length, lying/upright, residual, conf., grasp point; 14-D rel. to tip) + tip state, IK target, lift, arm q, seal/suction flags, tote offset, tool force, last action	5-D @30 Hz: tip Δxyz (1.2 cm), suction on/off, lift Δ; suction wand + tote on trunk	BC on 1.7k scripted demos (DART noise, 3.9 min on GB10): DDPM diffusion (2.2 M, chunk 8, 8 DDIM steps, 15 min) and MLP-BC (1.6 M, 5 min). DR: 8 can sizes, 0.1–0.5 kg, 65% lying	GB10 edge; MLP 0.16 ms, diffusion 12.2 ms	Isaac, 768 seeded scenarios: MLP-BC **91.0%**, diffusion 89.7%, scripted 90.0%; tool force max 8.7 N (MLP) vs 226 N (diffusion). VLM grasp point (Gemini Robotics-ER): 97% on target
5 Rust repair orchestrator `policy/orchestrator.py`	detector confidence, rust fraction, contact force, person distance, lens quality, gust force, rain	state: detect→2nd view→spray→soak 30 s→scrub (1/2) → verify → paint → verify paint	hand-written state machine; safety states BACK_OFF (F>40 N), HOLD, PAUSE_PERSON (<3 m), ESCALATE, WIPE_LENS (q<0.7), WAIT_LULL (gust >12 N)	GB10 edge, Python, ≪1 ms/decision	Isaac live run (colour-detector stand-in): 94.7% rust removed, scrub-phase peak 51 N for 8 ms then 1 back-off; ended ESCALATE (honest fail-safe). 120 glare scenarios: 25 unsafe, 17 of them faint patches (<0.65)
6 Gas-leak job + gas bot `policy/leak, gasbot`	trunk IR CH_4 (t_{90} 6 s), arm-tip catalytic (t_{90} 9 s), H_2S cell (t_{90} 25 s), ultrasonic level + bearing, OGI camera; flange views (torque stripes, crack check)	gas check→cast-and-surge seek→inspect→re-torque→verify; retreat / escalate	rules + estimators; plume = orifice eq. (choked, 3 mm/20 bar → 16 g/s) + Warp 3D advection–diffusion (368k cells, 4.2× real time)	GB10 edge; crack read 3.1 ms (TRT FP16)	Mock: 60/60 leaks detected, located, fixed; median 8 s detect, 5 s to source, 25 s fixed; 0 false alarms in 120 no-leak runs; without ultrasound 22/60. Cracks & big leaks: 24/24 escalated. Crack check 40/40, 0 false alarms
7 Navigation `policy/fleet, rigplanner`	walkway/stair graph of the rig, hazards (crates, closed walkways, gas zones, persons), weather (wind, wet, fog)	path + ETA	graph search with weather costs (storm: speed×0.63, stairs×0.40, windward walkways closed)	edge / fleet host, <1 ms	Judges scenario, 3 robots: 5 re-routes, crate drop +59 m/+97 s, 5 jobs done, 0 corridor conflicts; 2 jobs cut off by storm handed to crew
8 Fleet coordinator `fleet/`	robot poses/states, battery (432 Wh, dock), task risk priority, reservations, approvals	assignment, dock, approval tier AUTO / SUPV / ASSIST / HUMAN	Hungarian assignment (ETA + priority), corridor reservations, person-pause	fleet host, CPU	100 simulated shifts: 0 unsafe actions, 100/100 leaks isolated in 398 s mean, 88 jobs/h
9 Gemini 4 Argon control centre `policy/control`	SensorSnapshot: robot states, force, gas %LEL/ppm, acoustic, plume kg/h, verifier verdicts, weather, jobs, 1 image	advisory actions + rationale; never approves	prompted LLM (`gemini-4-argon`) every 30 s; edge rules always win (F>40 N, ≥50% LEL evacuate, person <3 m); envelope refusals; cut-link → edge only	cloud; 8.5 s median/call (measured), ≈$0.003/call in the scrub bench; cap $5/day	40 shifts, sim-verified plans (replay planner; live Gemini run cut for time): 0 unsafe, bad-plan leak isolation 437 → 406 s
10 Perception `detection/`	RGB 224×224 crop	rust prob. + patch map	MobileNetV4-small + LoRA r16 on real + sim crops	GB10 TRT FP16 0.35 ms p50	Real test (881): F1 0.977, AUROC 0.998; held-out sim renders 13/13. Argon as detector: F1 0.80, 9.5 s/image

Sensors → policy. • = used as input; ○ = indirect (via another policy / state).

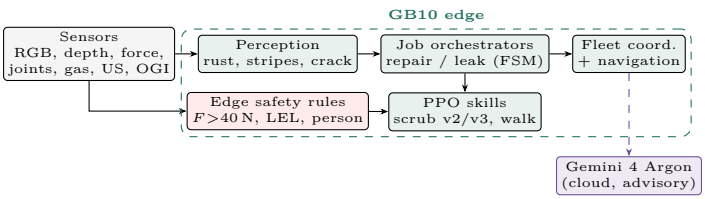
	RGB cam	depth/ lidar	con- tact F	joint propr.	gas IR/cat	H_2S	ultra- sound	OGI	wea- ther/IMU	fleet state
1 Scrub PPO v2	○		•	•						
2 Storm v3	○		•	•					○	
3 Walking		•		•					•	
4 Can pick-up	○	•	•	•						
5 Repair orch.	•		•						•	○
6 Gas bot / leak	•				•	•	•	•	•	
7 Navigation		○			○				•	•
8 Fleet coord.					○				○	•
9 Gemini Argon	•		•		•	•	•		•	•
10 Perception	•									

Scrub control: before RL vs after RL vs Gemini 4 Argon. Success = ≤5% rust left in the 15 s episode; "in 7.45 s" = within PPO's mean clean time.

		Isaac	CPU mock, same 30 seeded scenarios				
	controller	succ.	succ.	in 7.45 s	removed	clean s	decision	$/ep
pre-RL	random	0.0%	0.0%	0.0%	30.5%	—	0.045 ms	0
	greedy	32.0%	30.0%	0.0%	68.8%	12.5	0.081 ms	0
RL	PPO v2	**95.3%**	**93.3%**	**53.3%**	**94.5%**	**7.6**	**0.074 ms**	0
LLM	Argon, paused	—	36.7%	0.0%	83.0%	11.5	8.5 s	0.081
	Argon, real time	—	0.0%	0.0%	3.2%	—	8.0 s	0.006

Isaac: 256 ep (PPO 1.9% ep. >40 N vs greedy 46.5%, random 68.4%). Gemini gets a JSON state summary and the greedy action interface. Paused sim: 28 calls and 298 s wall per 15 s episode (≈20× slower than real time). Real time: the arm gets ≈2 decisions per episode. PPO decides ≈10^5× faster, free, on the robot. Total API spend $2.61 (`runs_sample/bench/GEMINI_RESULTS.md`).

Where it runs. Everything that moves the robot runs on the **NVIDIA GB10 edge box** (Grace–Blackwell, 20 Arm cores): detector 0.35/0.39 ms p50/p99 (TRT FP16, same decisions as FP32) + scrub policy 17 μs ⇒ a 2713 Hz loop ceiling vs. the 30 Hz the policy needs; with the GPU 94% busy rendering Isaac it still closes at 418 Hz (198 Hz p99). GPU draw 18 W serving the detector. No network is needed for any decision. **Cloud**: only Gemini 4 Argon (fleet-level advice) and an optional Gemini verifier second opinion; cutting the link drops to edge-only.



What's novel (opt-in upgrades, before/after in `runs_sample/novelty/`)

- **Scrub:** force is an RL action with a quadratic over-force cost: 76.8% → 1.0% episodes >40 N; an opt-in CBF force shield (RLS stiffness) takes calm >40 N 0.8% → 0% and storm 11.5% → 6.6% (not 0; storm worst peak 210 → 252 N).
- **Storm:** storm-randomised fine-tune cuts the worst-case contact force 274 N → 69 N at full storm.
- **Walking:** one gait for stowed *and* deployed arm (payload α in obs): 1.94 falls/km; opt-in RMA (1 s joint history) cuts velocity error on a heaving deck with 5 kg payload 0.077 → 0.063 m/s (−18%), falls unchanged (6.84 → 6.77/km).
- **Pick-up:** BC on depth-fitted cylinder geometry (no pixels) learns pick-up in 5–15 min (91.0% vs scripted 90.0%); Gemini Robotics-ER grasp point on unseen objects 7% → 97%.
- **Repair FSM:** soak-before-scrub, verify-before-paint, lens/gust/person safety states; 0% >40 N with the sensor in the loop.
- **Gas:** Bayesian particle filter + infotaxis finds the CH_4-only leak 64% → 86%, wrong-flange hand-offs 2 → 0.
- **Fleet / Argon:** plans are rehearsed in a fast sim before use: bad-plan leak isolation 437 → 406 s, 0 unsafe.
- **Perception:** shift-gated test-time adaptation: storm-lens F1 0.47 → 0.95, clear-weather F1 0.993 unchanged.

Honest caveats. (i) The Isaac live repair run uses a *colour stand-in* detector, not the LoRA model; the LoRA F1 0.977 is published in `detection/RESULTS.md`, not re-measured here (the edge build of the sim-trained detector scores 12/13 on held-out renders). (ii) Scrub legs are fixed in training; trunk sway only via the storm model. (iii) Walking and scrubbing are separate policies; some fleet demos move robots kinematically. RMA does not reduce falls; the CBF shield does not stop storm spikes. (iv) Gas results are a CPU mock with modelled sensor lags; the plume solver is validated against a Gaussian plume (reported ≤4.5% centreline error, 2–9 m) but not against field data. (v) Gemini scrub numbers are CPU-mock only (n=30); no Gemini action is ever executed without an edge rule and envelope check.